Q1.Explain Performance improvement in machine learning.

Ans: Performance improvement in machine learning means making a model more accurate and efficient so it can give better results on new data.

A model may not perform well due to problems like overfitting (too complex) or underfitting (too simple). So, different methods are used to improve its performance.

Methods to improve performance:

Data Preprocessing

Feature Selection

Choose Proper Algorithm

Hyperparameter Tuning

Regularization

Cross-Validation

Ensemble Methods

Q2.Difference between Predictive/Descriptive.

Ans:

Predictive Analysis	Descriptive Analysi
Predicts future outcomes	Describes past data
Focus on what will happen	Focus on what has happened
Uses historical data with patterns	Uses only historical data
Goal is forecasting	Goal is understanding data
Uses ML, regression techniques	Uses reports and summaries
Example: Predict next month sales	Example: Analyze past sales

Q3.Define Ensemble for Performance improvement

Ans: Ensemble is a machine learning technique in which multiple models are combined to improve the overall performance of prediction.

Instead of relying on a single model, ensemble methods take the outputs of several models and combine them to produce more accurate and reliable results.

Examples:

Bagging (Random Forest)

Boosting (AdaBoost)

Q4 . Describe Model representation and interpretability

Ans: Model Representation:

Model representation refers to how a machine learning model is structured and how it expresses the relationship between input and output. It defines the form of the model and how it makes predictions.

Example:

Linear Regression → equation

Decision Tree → tree

Interpretability:

Interpretability means how easily a human can understand the model's decisions or predictions. It shows whether we can explain why a model gave a particular output.

Examples:

Decision Trees are easy to interpret

Neural Networks are difficult to interpret

Q5.Write any three examples of supervised learning in the field of healthcare.

Ans: Examples of Supervised Learning in Healthcare

Disease Prediction

Predicting diseases like diabetes or heart disease using patient data.

Medical Image Diagnosis

Classifying X-rays or MRI scans to detect tumors or infections.

Patient Outcome Prediction

Predicting whether a patient will recover or need further treatment based on medical history.

Q6 .write any three applications of regression.

Ans: Applications of Regression

Sales Forecasting

Predicting future sales based on past data.

House Price Prediction

Estimating price of a house based on features like size and location.

Demand Prediction

Predicting product demand to manage inventory.

Q7.find a linear relationship between the hours smartphone used (input feature) and the exam score(output target)

Not available

Q8.Explain Support Vectors in the SVM model.

Ans: Support Vectors are the data points that lie closest to the decision boundary (hyperplane) in a Support Vector Machine (SVM).

They are important because they determine the position of the hyperplane and help in creating the maximum margin between different classes. Only these points directly affect the model.

Key Points:

Nearest points to the decision boundary

Define the hyperplane

Help maximize margin

Most important data points